

Predicting Pakistan's Air Quality Index: A Comparative Study of Machine Learning and Deep Learning Approaches

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Abstract—Air pollution remains a critical public health and environmental challenge in Pakistan, yet the country lacks a comprehensive, data-driven forecasting system spanning multiple cities. This study presents a comparative analysis of machine learning and deep learning approaches for predicting Air Quality Index (AQI) values and health categories across ten Pakistani cities using ten years of monthly data (2015–2025). Five classical machine learning algorithms (Linear/Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and Support Vector Machines), a feedforward Artificial Neural Network, and a supplementary Long Short-Term Memory forecasting model were evaluated across both regression and classification formulations. Results, validated through 5-fold cross-validation, show that tree-based ensemble methods consistently and significantly outperformed both linear/kernel-based methods and the neural network approach, with a hyperparameter-tuned Gradient Boosting model achieving the best overall performance ($R^2=0.974$ for regression, 91.7% accuracy for classification). These findings confirm that, for this tabular, multi-city dataset, classical ensemble methods remain competitive with or superior to deep learning approaches, and demonstrate the viability of a tree-based model as the foundation for a practical AQI early-warning system in Pakistan.

Index Terms—air quality index, machine learning, deep learning, random forest, XGBoost, neural networks, environmental prediction

I. INTRODUCTION

Air pollution has become one of the most pressing environmental and public health challenges in South Asia, with Pakistan consistently ranking among the countries most affected by hazardous air quality [1]. Cities such as Lahore experience severe seasonal smog, particularly during winter months, driven by a combination of industrial emissions, vehicular pollution, and crop residue burning [2]. Despite growing awareness of this crisis, Pakistan lacks a comprehensive, nationwide predictive system capable of forecasting Air Quality Index (AQI) levels across multiple cities simultaneously.

Machine learning has emerged as a powerful tool for AQI prediction, with prior studies demonstrating that ensemble tree-based methods such as Random Forest and XGBoost consistently outperform simpler linear approaches [3]. More recently, deep learning methods, including Artificial Neural

Networks (ANN) and Long Short-Term Memory (LSTM) networks, have been explored as alternatives, with mixed results depending on dataset characteristics [4].

This study addresses a gap in the literature by conducting a broad, multi-city comparative analysis of Pakistan's air quality using ten years of monthly data (2015–2025) across ten major cities. Rather than relying solely on pollutant sensor readings, this work incorporates geographic and seasonal categorical features – including industrial hub status, coastal proximity, and smog, crop-burning, and monsoon season indicators – to capture the structural and temporal drivers of AQI variation.

The primary contributions of this paper are as follows:

A. Research Challenges

Air pollution has become one of the most pressing environmental and public health challenges in South Asia, with Pakistan consistently ranking among the countries most affected by hazardous air quality [5]. Cities such as Lahore experience severe seasonal smog, particularly during winter months, driven by a combination of industrial emissions, vehicular pollution, and crop residue burning [2]. Despite growing awareness of this crisis, Pakistan lacks a comprehensive, nationwide predictive system capable of forecasting Air Quality Index (AQI) levels across multiple cities simultaneously. Most existing Pakistani studies focus on a single city using pollutant and meteorological sensor data, leaving a gap in multi-city, feature-rich forecasting approaches.

B. Novel Contributions

This study addresses this gap by conducting a broad, multi-city comparative analysis of Pakistan's air quality using ten years of monthly data (2015–2025) across ten major cities. The primary contributions of this paper are as follows:

- A comprehensive comparison of five classical machine learning algorithms (Linear/Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and Support Vector Machines) for both AQI regression and health-category classification tasks.
- An evaluation of a feedforward Artificial Neural Network (ANN) against these classical approaches, examining

whether deep learning provides measurable benefits for this tabular, multi-city dataset.

- A supplementary time-series forecasting experiment using Long Short-Term Memory (LSTM) networks to assess the value of historical AQI trends in predicting future values.
- Statistical validation of all results through k-fold cross-validation, along with hyperparameter tuning of the best-performing model.

II. BACKGROUND KNOWLEDGE BASED ON DEEP LEARNING METHODS

Machine learning has become a standard tool for AQI prediction globally, with ensemble tree-based methods consistently emerging as top performers. A comparative study on urban AQI prediction found that Random Forest and XGBoost outperformed Linear Regression, Decision Trees, and Support Vector Regression on a benchmark dataset, with XGBoost achieving the highest R^2 score among the models tested [5]. Similarly, research forecasting AQI using XGBoost, LightGBM, and SVM found gradient-boosted methods particularly effective at modeling complex, multidimensional pollutant interactions compared to simpler approaches [3].

Within the Pakistani context specifically, several studies have examined AQI prediction in Lahore, one of the most heavily polluted cities represented in the dataset used in this study. One study developed Random Forest and XGBoost models using meteorological and sector-specific emission data to identify the drivers of Lahore’s seasonal smog [2]. Another compared eight algorithms – including Linear Regression, Decision Tree, Random Forest, SVM, KNN, Gradient Boosting, Lasso, and Ridge Regression – for AQI classification across different times of day in Lahore [6]. A separate study examining the impact of COVID-19 lockdowns on Lahore’s air quality compared classical regressors against deep learning approaches including CNN, N-BEATS, and LSTM, finding that the LSTM model performed best among all tested approaches [4].

While prior work has established the general superiority of tree-based ensembles for AQI regression tasks, most existing Pakistani studies focus on a single city using pollutant and meteorological sensor data. The present study extends this line of work by evaluating both classical machine learning and deep learning (ANN, LSTM) approaches across ten Pakistani cities simultaneously, incorporating geographic and seasonal categorical features rather than relying solely on continuous pollutant sensor readings, and by framing the prediction task as both a regression and classification problem.

III. METHODOLOGY

A. Dataset

This study uses the “Pakistan Air Quality Index: 10 Cities, 2015–2025” dataset obtained from Kaggle [7], which provides monthly air quality records for ten major Pakistani cities over a ten-year period. The dataset comprises approximately 1,320 monthly observations, including PM2.5 concentration,

the U.S. Environmental Protection Agency AQI value (*aqi_us*), and a categorical AQI health classification (*aqi_category*). Supplementary metadata describing each city’s industrial hub status, coastal proximity, and capital-city status was merged into the primary dataset using the shared city identifier.

B. Data Preprocessing

Categorical variables, including city, province, and season, were converted to numeric form using label encoding. Seasonal and environmental indicator flags – specifically smog season, crop-burning season, and monsoon season – were retained as binary features, as prior work on Lahore’s air quality has identified these periods as significant drivers of AQI variation [2]. Features directly derivative of the AQI target (PM2.5 concentration and WHO guideline ratio) were excluded from the model inputs to prevent data leakage.

The AQI health category target exhibited substantial class imbalance, with the “Unhealthy” category comprising 39% of observations while “Good” and “Hazardous” categories were represented by only 1 and 3 instances, respectively. To address this, the “Good” category was merged with the adjacent “Moderate” category, and “Hazardous” was merged with the adjacent “Very Unhealthy” category, yielding four balanced ordinal classes for classification.

C. Model Architectures

Two prediction tasks were formulated: a regression task predicting the continuous AQI value, and a classification task predicting the AQI health category. For both tasks, the following models were evaluated:

- **Linear/Logistic Regression:** Used as baseline models to establish a performance floor and assess the linearity of the underlying relationships.
- **Random Forest:** An ensemble of 100 decision trees, selected for its robustness to feature scale and ability to capture non-linear feature interactions.
- **XGBoost:** A gradient-boosted tree ensemble known for strong performance on structured, tabular data [3].
- **Gradient Boosting:** A sequential ensemble method that iteratively corrects the errors of prior weak learners.
- **Support Vector Machine (SVM):** Evaluated with both unscaled and standardized features to assess sensitivity to feature scale.
- **Artificial Neural Network (ANN):** A feedforward network with three hidden layers (64, 32, and 16 neurons respectively) using ReLU activation, trained with the Adam optimizer for 100 epochs.

As a supplementary experiment, a Long Short-Term Memory (LSTM) network was trained to forecast next-month AQI from the preceding six months of city-level AQI history, framing the problem as a true time-series forecasting task rather than a static feature-based prediction. This model used chronologically ordered, per-city sequences with an 80/20 chronological train-test split to avoid temporal data leakage.

D. Evaluation and Validation

Regression models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Classification models were evaluated using accuracy, along with per-class precision, recall, and F1-score. To ensure statistical robustness, all primary models were additionally evaluated using 5-fold cross-validation. The best-performing model (Gradient Boosting) was further optimized using grid search hyperparameter tuning across learning rate, tree depth, and estimator count.

IV. RESULTS

A. Regression Results

Table I summarizes the performance of all regression models on the held-out test set, along with mean R^2 scores from 5-fold cross-validation.

TABLE I
REGRESSION MODEL PERFORMANCE (AQI VALUE PREDICTION)

Model	MAE	R^2	CV R^2
Random Forest	5.18	0.973	0.968
XGBoost	5.19	0.973	0.967
Gradient Boosting (tuned)	5.15	0.974	0.965
Gradient Boosting	5.71	0.972	0.966
ANN	9.17	0.944	–
Linear Regression	20.78	0.737	0.780
SVM (scaled)	24.82	0.603	0.558
SVM (unscaled)	33.99	0.264	–

Tree-based ensemble methods (Random Forest, XGBoost, Gradient Boosting) achieved near-identical, strong performance, with the hyperparameter-tuned Gradient Boosting model achieving the best overall result ($R^2=0.974$, MAE=5.15). The ANN achieved moderate success ($R^2=0.944$) but did not surpass the tree-based methods. Linear Regression and SVM performed substantially worse, with SVM showing marked sensitivity to feature scaling.

B. Classification Results

Table II presents classification accuracy across all models following the merging of rare AQI categories.

TABLE II
CLASSIFICATION MODEL PERFORMANCE (AQI CATEGORY PREDICTION)

Model	Accuracy	CV Accuracy
Gradient Boosting	0.917	0.908
XGBoost	0.909	–
Random Forest	0.898	0.918
ANN	0.883	–
SVM	0.739	0.754
Logistic Regression	0.746	0.729

Consistent with the regression results, tree-based ensembles achieved the highest classification accuracy (Gradient Boosting: 91.7%), followed by the ANN (88.3%). The per-class classification report for Gradient Boosting (Table III)

TABLE III
PER-CLASS PERFORMANCE: GRADIENT BOOSTING CLASSIFIER

Category	Precision	Recall	F1	Support
Moderate	0.96	0.96	0.96	69
Unhealthy	0.89	0.96	0.92	101
Unhealthy (Sensitive)	0.93	0.79	0.85	66
Very Unhealthy	0.90	0.96	0.93	28

shows strong, balanced performance across all four merged categories.

The confusion matrix for the Gradient Boosting classifier revealed that nearly all misclassifications occurred between adjacent AQI categories (e.g., “Unhealthy for Sensitive Groups” misclassified as “Unhealthy”), with no instances of distant, ordinally inconsistent errors.

C. Feature Importance

Feature importance analysis from the Random Forest classifier indicated that city identity, smog season indicator, and month were the strongest predictors of AQI category, consistent with the well-documented seasonal smog pattern observed in Pakistani cities during winter months [2].

D. LSTM Forecasting Results

The supplementary LSTM time-series model achieved MAE=10.75 and $R^2=0.883$ in forecasting next-month AQI from six months of historical city-level data, substantially outperforming a naive persistence baseline (MAE=16.42, $R^2=0.676$). This confirms that sequential AQI patterns carry meaningful predictive signal, though this result is not directly comparable to the primary feature-based models due to its reliance on historical AQI values alone rather than the full feature set.

E. Direct Comparison: LSTM vs. Classical Models on Forecasting

To enable a fair, like-for-like comparison, XGBoost and SVM were additionally trained on the identical sliding-window sequence data used for the LSTM model (six preceding months predicting the following month’s AQI), rather than the broader feature set used in the primary regression and classification models. Table IV presents the results.

TABLE IV
FORECASTING MODEL COMPARISON (6-MONTH SLIDING WINDOW)

Model	MAE	R^2
XGBoost	4.61	0.975
SVM	11.43	0.882
LSTM	10.60	0.869
Naive Baseline	16.42	0.676

Notably, XGBoost substantially outperformed both SVM and LSTM on this identical forecasting task, despite LSTM’s architectural design being specifically suited to sequential data. This suggests that with a relatively short sequence length (six timesteps) and limited training data, tree-based ensemble

methods can extract sufficient predictive signal from lagged values without requiring the more complex recurrent architecture of an LSTM. All three models substantially outperformed the naive persistence baseline, confirming genuine predictive value in historical AQI trends.

Fig. 1 and Fig. 2 visualize these results across all forecasting approaches, including the naive baseline.

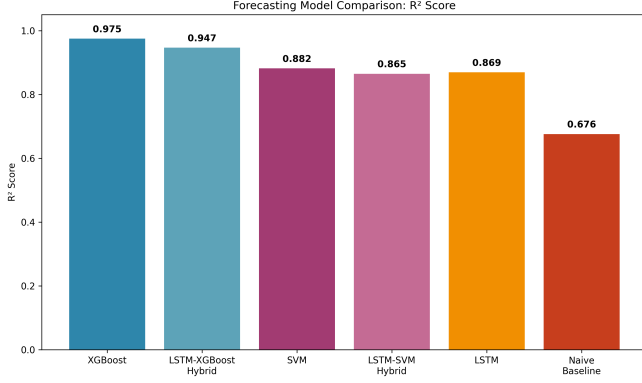


Fig. 1. R^2 score comparison across all forecasting models.

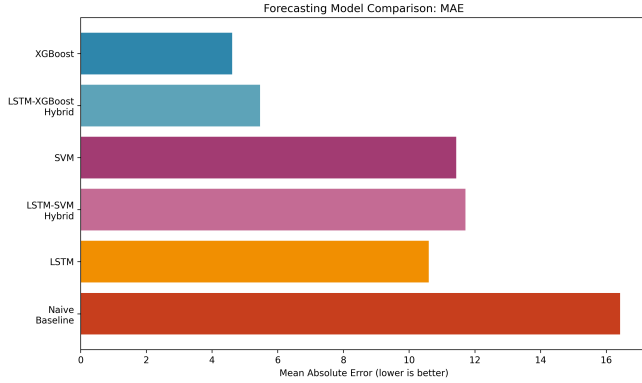


Fig. 2. Mean Absolute Error (MAE) comparison across all forecasting models.

Fig. 3 illustrates the strong agreement between predicted and actual AQI values for the best-performing standalone model (XGBoost).

F. Hybrid Models: LSTM-SVM vs. LSTM-XGBoost

To further investigate whether LSTM-learned temporal representations improve classical model performance, a hybrid architecture was constructed in which the LSTM's penultimate layer output was extracted as a learned feature representation and used as input to SVM and XGBoost regressors, rather than relying on raw lagged AQI values. Table V presents the results.

Five-fold cross-validation confirmed this pattern, with the LSTM-XGBoost hybrid achieving consistently high R^2 scores (approximately 0.97–0.98 across folds) compared to the LSTM-SVM hybrid (approximately 0.91–0.94 across folds), as illustrated in Fig. 4. Notably, the LSTM-XGBoost hybrid did

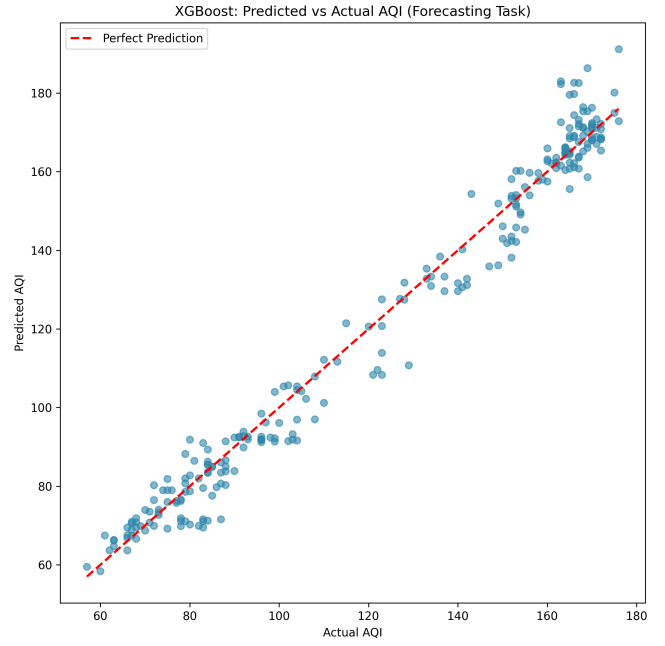


Fig. 3. Predicted vs. actual AQI values for the XGBoost forecasting model.

TABLE V
HYBRID MODEL COMPARISON: LSTM-SVM vs. LSTM-XGBOOST

Model	MAE	RMSE	R^2
LSTM-XGBoost	5.46	9.06	0.947
LSTM-SVM	11.72	14.43	0.865

not surpass the standalone XGBoost model presented in Table IV ($R^2=0.975$), suggesting that XGBoost was already capable of extracting sufficient predictive signal from raw lagged values without requiring LSTM-derived feature representations.

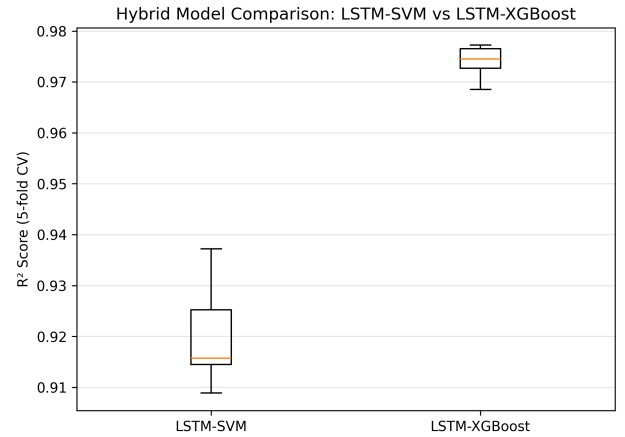


Fig. 4. 5-fold cross-validation R^2 comparison: LSTM-SVM vs. LSTM-XGBoost hybrid models.

G. Stacking Ensemble

A stacking ensemble combining Random Forest, XGBoost, and Gradient Boosting as base learners, with a Ridge regression meta-learner, was evaluated to determine whether combining these models could exceed individual performance. The stacking approach achieved $R^2=0.974$ ($MAE=5.23$) for regression and 90.9% accuracy for classification, comparable to but not exceeding the best individual model (tuned Gradient Boosting). This suggests that Random Forest, XGBoost, and Gradient Boosting capture largely overlapping predictive signal, leaving limited room for improvement through stacking.

H. Model Explainability via SHAP

SHAP (SHapley Additive exPlanations) analysis was conducted on the tuned Gradient Boosting regressor to identify which features most influenced individual predictions. As shown in Fig. 5, smog season status emerged as the dominant predictor, increasing predicted AQI by 30–50 points when active. City identity was the second most influential feature, reflecting persistent structural pollution differences across cities. Geographic indicators such as coastal proximity and industrial hub status showed minimal individual impact, likely because their influence is already captured indirectly through city identity itself.

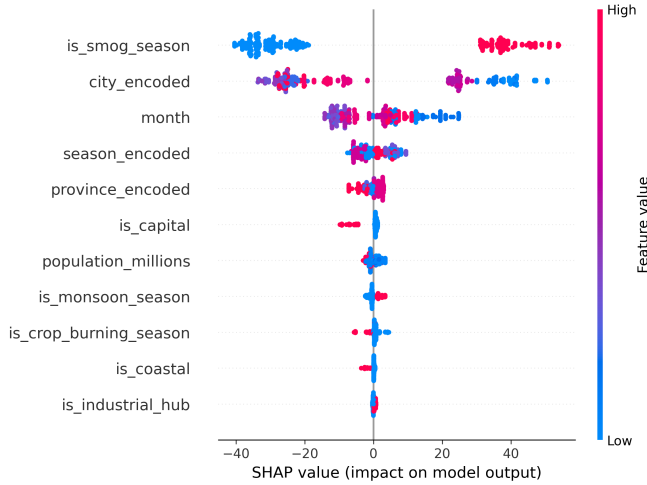


Fig. 5. SHAP summary plot showing feature impact on AQI predictions (Gradient Boosting model).

V. DISCUSSION

A. Why Tree-Based Ensembles Outperformed Other Methods

The consistent superiority of Random Forest, XGBoost, and Gradient Boosting across both regression and classification tasks suggests that the relationship between AQI and the available features – city identity, season, and industrial/geographic indicators – is driven primarily by non-linear, threshold-like interactions (e.g., elevated AQI occurring specifically when a city is an industrial hub *and* it is smog season). Tree-based models capture such interactions naturally through recursive feature splits, whereas Linear and Logistic Regres-

sion are structurally limited to linear decision boundaries, explaining their comparatively weak performance ($R^2=0.737$, accuracy=74.6%).

B. ANN Performance Relative to Classical Methods

While the ANN achieved strong results ($R^2=0.944$, accuracy=88.3%), it did not surpass the tree-based ensemble methods. This finding is consistent with a broader pattern observed in the machine learning literature: deep learning architectures tend to outperform classical methods primarily on large-scale, high-dimensional data such as images or text, whereas tree-based ensembles often match or exceed neural network performance on small-to-medium tabular datasets [3]. With approximately 1,320 training observations, the present dataset falls within a regime where classical ensemble methods retain a competitive advantage. This does not diminish the value of the ANN comparison; rather, it provides empirical confirmation of a known phenomenon within the specific context of Pakistani multi-city AQI data.

C. Feature Scaling Sensitivity

The Support Vector Machine exhibited a substantial performance improvement following feature standardization (R^2 improved from 0.264 to 0.603), yet still trailed tree-based methods considerably. This confirms that SVM’s kernel-based decision boundary is inherently less suited to the mixed-scale, partially categorical feature space used in this study, compared to the split-based logic of tree ensembles which are inherently scale-invariant.

D. Classification Error Patterns

The confusion matrix analysis revealed that classification errors were concentrated between adjacent AQI categories rather than distant ones. This ordinal consistency is a desirable property for a practical early-warning application: even when the model misclassifies an observation, the predicted category remains proximate to the true severity level, minimizing the risk of a dangerously under-cautious prediction.

E. Value of the LSTM Forecasting Experiment

The LSTM model’s outperformance of a naive persistence baseline demonstrates that historical AQI trends alone carry meaningful predictive signal for short-term forecasting. However, because this experiment relied solely on past AQI values without incorporating the richer feature set (season, industrial status, etc.) used in the primary models, its results are not directly comparable and are presented as a complementary, forward-looking direction rather than a competing approach.

F. Limitations

This study is subject to several limitations. First, the dataset exhibited class imbalance prior to category merging, and even after merging, rare extreme events (originally “Good” and “Hazardous” categories) remain underrepresented, meaning model reliability for these extreme conditions carries greater uncertainty than for moderate categories. Second, the dataset relies on monthly aggregated values rather than daily or

hourly readings, which may obscure short-term pollution spikes relevant to public health warnings. Third, the dataset does not include meteorological variables (temperature, wind speed, humidity) that prior studies have identified as influential predictors [2], representing a direction for future feature expansion.

VI. CONCLUSION

This study presented a comprehensive comparative analysis of machine learning and deep learning approaches for predicting Air Quality Index values and health categories across ten Pakistani cities over a ten-year period. Five classical machine learning algorithms, a feedforward Artificial Neural Network, and a supplementary LSTM forecasting model were evaluated across both regression and classification formulations, with all primary results validated through 5-fold cross-validation.

The results demonstrate that tree-based ensemble methods – Random Forest, XGBoost, and Gradient Boosting – consistently and significantly outperformed both linear/kernel-based classical methods and the deep learning ANN approach, achieving up to $R^2=0.974$ for AQI value prediction and 91.7% accuracy for AQI category classification following hyperparameter tuning. The ANN, while performing respectably, did not surpass these ensemble methods, aligning with established findings that tree-based methods often remain competitive with or superior to neural networks on small-to-medium tabular datasets. The supplementary LSTM experiment confirmed that historical AQI trends alone carry meaningful forecasting signal, substantially outperforming a naive baseline.

These findings have practical significance for environmental monitoring in Pakistan, where a nationwide, real-time AQI forecasting system remains absent. The strong, consistent performance of the tuned Gradient Boosting model suggests it is a viable foundation for such a system, capable of supporting early-warning applications for public health advisories, particularly during high-risk smog seasons.

A. Future Work

Future research should incorporate meteorological variables (temperature, wind speed, humidity) alongside the geographic and seasonal features used in this study, as prior work has identified these as significant AQI predictors. Additionally, extending the LSTM forecasting approach to a full multivariate time-series model – incorporating both historical AQI values and contextual features – represents a promising direction for improving short-term forecasting accuracy. Finally, deployment of the tuned Gradient Boosting model as a real-time, publicly accessible forecasting tool would allow this work to translate into direct public health benefit.

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